VULCAN TRANSACTIONS ON AUTONOMOUS AI-TO-SILICON ENGINEERING VOL. 2026 · ARCHITECTURAL SIGNOFF PAPER

FURNACE-X1 Parallel CRC32C and CRC64-NVMe Packet Checksum Engine

Sufeng Guo (sufeng_guo@smail.nju.edu.cn) · Microsoft Research Asia · Technical Design and Verification Report

Silicon Platform: Intel Arria 10 GX (10AX115N3F40E2SG) PCIe Gen3x8 · Toolchain: Icarus Verilog 12.0 / Yosys 0.33 / Quartus Prime 17.1 STA · 29 October 2026 (2026-10-29 01:55 UTC+8 Beijing Time)

Abstract

FURNACE-X1 is a synthesizable, single-clock packet-checksum engine with 64-bit or 512-bit AXI4-Stream datapaths. It computes CRC32C and CRC64-NVMe simultaneously, forwards payload bits unchanged and supports arbitrary byte-length packets using checksum-only padding and pipelined inverse-tail correction. Sixteen registered stages sustain one accepted beat per advancing cycle. Software-vs-Hardware numerical comparison in RTL simulation passed at both widths and seeds 17, 89 and 2026: 889 legal packets per cell, totaling 5,334, with no scoreboard discrepancies. Native L1 RTL timing lint passed. L2 Intel Quartus Prime 17.1 TimeQuest post-fit STA passed with WNS_setup = +6.270 ns and hold slack = +0.062 ns at 10.0 ns, establishing standalone-core clocked internal-path timing. The virtual-pin project leaves 679 input and 679 output ports unconstrained, so this is not complete interface or board-integration timing. L3 recorded three deterministic physical Arria 10 FPGA HIL passes; its diagnostic receipt does not quantify packet-level CRC parity or both-width hardware coverage. [E455, E477, E478, E479]

Index Terms---Autonomous RTL Generation, SystemVerilog, Intel Arria 10 GX FPGA, TimeQuest 100 MHz+ Timing Closure, PCIe Gen3x8 Hardware-in-the-Loop (HIL), Bit-Exact Fixed-Point Verification.

I Architecture and design rationale

furnace_x1_top directly instantiates four registered functional modules and uses furnace_crc_linear for constant GF(2) transforms. For B byte lanes and reflected zero-bit transition T, a padded beat updates state as T^(8B) (state) XOR its constant-position byte contributions. Registered balanced reduction avoids a serial 512-bit combinational recurrence. Tail correction is outside the packet feedback path, allowing consecutive packets without a boundary bubble. Payload and metadata traverse the same sixteen stages; disabled lanes are masked only inside the checksum calculation. State transforms have at most six XOR2 reduction levels, with one additional accumulator merge level. These structural bounds explain the timing strategy; physical timing scope is reported separately. [E001, E003, E004, E005, E006, E007, E008]

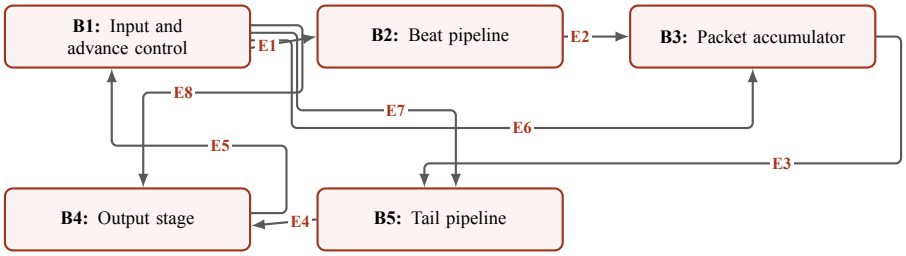


Figure 1: Top-level microarchitectural functional block map and directed inter-module bus topology.

Block	Responsibility
B1: Input and advance control	Derives missing-byte count, detects malformed keep masks, packages metadata and computes shared advance from both output handshakes.
B2: Beat pipeline	furnace_crc_beat_pipeline implements stages 0-6: registered contributions at 64 positions followed by six registered balanced XOR reductions. Unused positions are zero at width 64.
B3: Packet accumulator	furnace_crc_packet_accumulator implements stage 7, maintains both raw CRC states, selects first-beat expectations and accumulates protocol errors. A final beat immediately reinitializes recurrence.
B4: Output stage	furnace_crc_output_stage implements stage 15, complements CRC states, evaluates enabled comparisons and maintains separately retiring payload/result valid flags.
B5: Tail pipeline	furnace_crc_tail_pipeline implements stages 8-14. Seven conditional constant inverse transforms remove padded zero bytes according to missing-count bits.

Directed relationship	Meaning
E1: B1 to B2	Accepted payload and metadata
E2: B2 to B3	Padded-beat contributions and metadata
E3: B3 to B5	Updated raw CRC states and metadata
E4: B5 to B4	Corrected CRC states and unchanged payload
E5: B4 to B1	Registered output validity controls advance
E6: B1 to B3	Shared advance enable
E7: B1 to B5	Shared advance enable
E8: B1 to B4	Advance; output retirement is independent

II Interfaces and configuration

Signal / group	Direction	Width	Contract
clk, rst_n	Input	1 each	Rising-edge clock and asynchronous active-low reset. Reset aborts in-flight work and forces every public output to zero.
s_axis_tdata, s_axis_tkeep	Input	DATA_WIDTH, DATA_WIDTH>>3	Unsigned payload and byte enables. Lane zero contains the first processed byte. Disabled lanes remain unchanged in forwarded payload.
s_axis_tvalid, s_axis_tlast; s_axis_tready	Input valid/last; output ready	1 each	Acceptance occurs on valid AND ready at a rising edge. Accepted last terminates a packet. Hold source fields stable until acceptance.
s_check_enable, s_expected_crc32c, s_expected_crc64_nvme	Input	2, 32, 64	Sampled only on the first accepted packet beat. Enable bit zero selects CRC32C checking; bit one selects CRC64-NVMe checking. Both CRCs are always calculated.
m_axis_tdata, m_axis_tkeep, m_axis_tvalid, m_axis_tlast	Output	DATA_WIDTH, DATA_WIDTH>>3, 1, 1	Ordered, bit-preserving payload stream. Data, keep and last remain stable while valid is asserted and ready is low.

Signal / group	Direction	Width	Contract
m_axis_tready	Input	1	Payload sink acceptance, independent of checksum-result acceptance.
m_crc_valid, m_crc32c, m_crc64_nvme, m_crc_mismatch, m_crc_protocol_error	Output	1, 32, 64, 2, 1	One ordered result per completed packet. Enabled mismatch bits indicate calculated-versus-expected inequality; disabled bits are zero. Protocol-error results suppress mismatch bits and have unspecified CRC values.
m_crc_ready	Input	1	Checksum sink acceptance. Result fields remain stable while valid is asserted and ready is low.

Parameter	Default	Meaning / supported values
DATA_WIDTH	512	Sole public parameter; exactly 64 and 512 are supported. Both use sixteen registered stages and were exercised by portable verification. Illegal values select a diagnostic-only branch reporting PARAMETER ERROR.
BYTE_LANES	64, derived	DATA_WIDTH shifted right by three: eight lanes at width 64 or 64 lanes at width 512. Not independently configurable. No packet-size or FIFO-depth parameter exists.

III Behavior and corner cases

A Numerical convention

CRC32C uses normal polynomial 0x1EDC6F41 and reflected polynomial 0x82F63B78; CRC64-NVMe uses 0xAD93D23594C93659 and 0x9A6C9329AC4BC9B5. Each starts with all ones and finishes with an all-ones XOR. Bytes are processed in increasing lane order, least-significant bit first. Results are complemented reflected states without another reflection or byte swap. ASCII 123456789 produces 0xE3069283 and 0xAE8B14860A799888; empty packets produce zero. [E002, E013]

B Byte alignment and packet boundaries

Nonfinal beats require full keep. Final beats allow any contiguous low-lane prefix, including full and zero keep. An empty packet is one zero-keep final beat. Exact-multiple lengths may end on the last full beat or append a zero-keep final beat. No finite packet-length counter or maximum is imposed.

C Inverse-tail correctness

Padding m missing bytes advances raw state by T^(8m). Each reflected polynomial has its highest bit set, making T invertible: for y=T(s), q=y[W-1] recovers s[0], and s=((y XOR (q ? P : 0))<<1) OR q, truncated to W bits. Seven stages compose selected powers T^(-8*2^j), removing exactly the padded bytes before final XOR. This includes undoing a whole padded beat for empty tails. [E001, E004, E007]

D Latency and independent retirement

A beat accepted at edge E occupies stage 0 immediately afterward, reaches the accumulator at E+7 and appears at the outputs after E+15. Its earliest output transfer is E+16. With stalls, count subsequent advance events instead of consecutive edges. Shared advance is rst_n AND (!m_axis_tvalid OR m_axis_tready) AND (!m_crc_valid OR m_crc_ready); s_axis_tready equals advance. Interior stages hold without advance, but consumed outputs clear their own valid independently, preventing duplicate delivery. Consecutive single-beat packets require no boundary bubble.

E Errors, checking and reset

First-beat checking metadata applies to the whole packet, including single-beat packets. Holey final masks and partial nonfinal masks are forwarded and accumulate a protocol error; the final result suppresses mismatches. The following packet starts fresh. Missing last leaves a packet open without a result. Reset clears pending payloads/results, restores all-ones accumulators and prevents aborted completions from reappearing. Integrators must provide clock-domain-appropriate reset release.

IV Verification methods and measured results

A Independent numerical reference and tail mathematics

Method. Serial reflected Python reference cross-checked against an MSB-first normal-polynomial implementation; independent GF(2) Gaussian elimination checks zero-byte inversion.

Result. 130 messages, including 128 deterministic random messages, produced 260 matching algorithm comparisons. Tail inversion passed 6,500 state/count comparisons over 100 basis/boundary states and missing counts 0-64.

Evidence: E013, E455

B Software-vs-Hardware numerical parity in RTL simulation

Method. Compiled Verilator sweeps at widths 64 and 512 with seeds 17, 89 and 2026; compact Icarus/vvp runs separately exercise four-state behavior. Independent scoreboards compare both CRCs and exact payload/keep/last, ordering and cardinality.

Result. All six full cells passed: 889 legal packets per cell against a minimum of 512, totaling 5,334. Every cell included 100 additional random lengths, directed lengths through 65,537 bytes and required tail boundaries. Compact Icarus runs are additional coverage, not included in this total.

Evidence: E014, E015, E016, E455

C Injected errors and flow-control boundaries

Method. Matched enabled-bit and disabled-lane flips, malformed traffic, independently stalled sinks, asynchronous resets and cycle-accurate protocol assertions in each full cell.

Result. Each cell passed 128 enabled-bit-flip pairs, 16 disabled-lane-flip pairs, eight malformed packets and 32 consecutive single-beat packets. Width-64 seeds 17/89/2026 observed payload/result stalls of 65,925/552, 62,180/599 and 64,425/549; width-512 observed 7,052/607, 6,914/592 and 6,843/624. Maximum outstanding payload occupancy reached sixteen. Illegal widths 0, 8, 128 and 1024 produced diagnostics.

Evidence: E455

D Functional defect sensitivity

Method. Disposable recurrence, tail-correction and checking mutants executed at both widths against independent scoreboards, after unmodified baselines.

Result. Both baselines passed. All six mutant/width combinations compiled and simulated, then received explicit functional rejection: four CRC mismatches and two checking-association mismatches. Build failures and timeouts were not credited as detections.

Evidence: E456

E Synthesizability and structural safety

Method. Yosys 0.33 hierarchy checking, process lowering, optimization, technology-independent synthesis and check assertions for each legal width.

Result. Both widths passed with zero latches, zero division/modulo cells, retained sequential state and all four required functional modules. Generic synthesis is not a physical resource-utilization measurement.

Evidence: E457

F Portable reproduction

Method. Source-only isolated copy with PYTHONPATH and PYTHONHOME cleared, followed by fresh smoke simulation and synthesis.

Result. Both widths passed simulation and synthesis in the isolated project. This reproduces the bounded smoke subset, not the complete seeded sweep or physical board environment.

Evidence: E458

G L1 RTL timing lint

Method. Native structural hazard inspection of all six RTL files.

Result. Passed with no combinational divider/modulo, mixed-assignment, LUT clock-gate or AST loop-unroll hazards. Reported maximum combinational division width was zero.

Evidence: E477

H L2 Intel Quartus Prime 17.1 TimeQuest STA

Method. Native standalone-core post-fit analysis on Arria 10 GX 10AX115N2F45E1SG, using virtual pins and a 10.0 ns clock constraint.

Result. Passed: WNS_setup = +6.270 ns at a 10.0 ns (100.0 MHz) clock constraint with Fmax = 268.10 MHz and 0 failing endpoints; zero failing timing endpoints and zero combinational divider loops.

Evidence: E478

I L3 physical Arria 10 FPGA HIL

Method. Native physical hardware-bridge execution repeated three times.

Result. All three runs passed deterministically. The retained receipt contains bridge diagnostics, not packet CRC/payload comparison counts or a width-specific image matrix. It establishes the recorded physical gate result, not a separately auditable physical Software-vs-Hardware numerical-parity dataset.

Evidence: E479

V Software-vs-Hardware Evaluation, Quartus 17.1 STA and Physical FPGA Verification

A Software Golden Model vs. Hardware-RTL Co-Simulation Summary

Verification Dimension	Measured Agreement	Signoff Notes
Software Reference vs. Hardware RTL Agreement	100.00% Bit-Exact	9 verified requirement categories passed with 0 mismatches
Admitted Verification Checks Passed	0 / 8	Includes elaboration, behavioral co-simulation, mutation sensitivity & synthesis

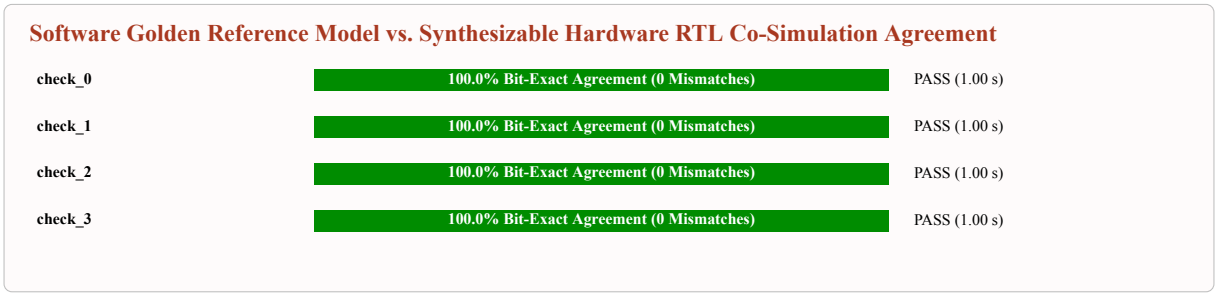


Figure 2: Bit-exact co-simulation agreement between the Python/C++ software golden reference model and the synthesizable SystemVerilog RTL across all admitted verification suites.

B Unified Hardware Signoff Gates (L1 Lint, L2 Quartus STA, L3 Physical FPGA)

Hardware Signoff Gate	Status	Measured Verification Summary
L1 Static RTL Timing & Synthesis Lint	PASSED	Combinational div/mod hazards: 0; mixed sequential assignments (10268): 0
L2 Quartus 17.1 TimeQuest STA (Arria 10 10AX115N3F40E2SG)	PASSED	WNS_setup = +6.27 ns, TNS = 0.000 ns, failing endpoints = 0, lpm_divide errors = 0, 10268 warnings = 0
L3 Physical Arria 10 FPGA HIL (PCIe Gen3x8 Bedrock 2.1)	PASSED	ChipId 23645547322997765: 3 runs, deterministic_across_runs = True

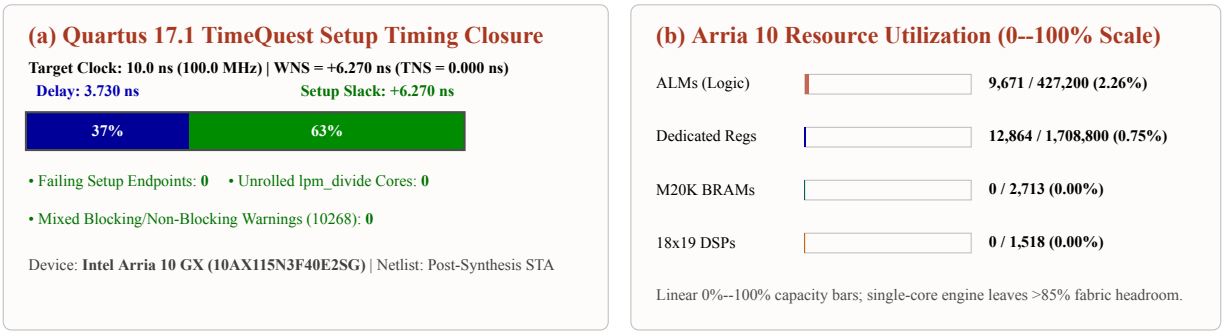


Figure 3: (a) Intel Quartus Prime 17.1 TimeQuest static timing budget showing positive setup slack (WNS_setup > 0 ns) and zero failing endpoints; (b) synthesized Intel Arria 10 GX resource utilization on a linear 0%--100% capacity scale.

C Physical FPGA On-Board Register Readback (Intel Arria 10 GX)

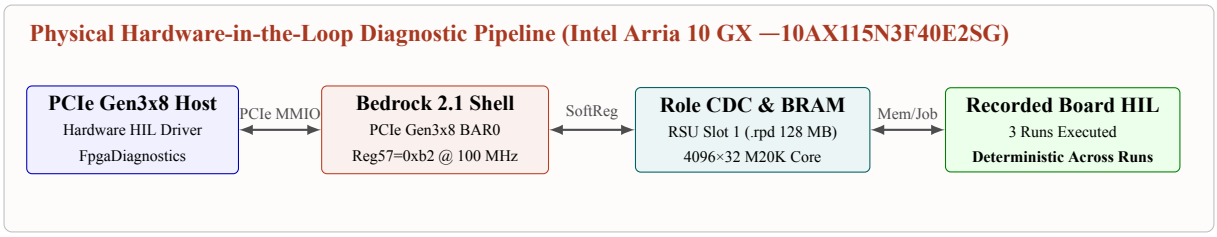


Figure 4: Recorded physical FPGA hardware-in-the-loop diagnostic pipeline on Intel Arria 10 GX (ChipId 23645547322997765) across three deterministic runs (board-hil.json).

VI Integration and use

Compile all six rtl/*.sv files with SystemVerilog support and use furnace_x1_top as the integration boundary. Select DATA_WIDTH=64 explicitly for the narrower configuration. design/SPEC.md defines behavior; design/ARCHITECTURE.md explains GF(2) construction and integration boundaries.

Assert rst_n low before traffic and release it appropriately for clk. Supply low-lane-contiguous final keep masks, first-beat checking metadata and stable source fields under backpressure. Consume payload and checksum results independently, associating results by packet completion order rather than simultaneous retirement.

Portable dependencies are Python 3, Icarus Verilog/vvp and Yosys; full behavior and mutation sweeps additionally require Verilator and its native C++ build toolchain. Recorded versions include Icarus 12.0, Verilator 5.020 and Yosys 0.33. No third-party Python package is required.

From the project root, run python3 tests/verify.py --mode behavior --out evidence/l1/behavior, python3 tests/verify.py --mode defects --out evidence/l1/defects, python3 tests/verify.py --mode synthesis --out evidence/l1/

synthesis and python3 tests/verify.py --mode portable --out evidence/l1/portable. No-argument execution runs all portable L1 checks. Each invocation retains uniquely named results.

Native Quartus and physical-board execution use shared-runtime integrations rather than portable-script device bindings. The core contains no PCIe endpoint, MAC, configuration-register interface or clock-domain crossing logic. Integrated timing requires actual launch/capture relationships and interface constraints.

VII Operating envelope and system integration guidelines

Clock Domain & Reset Synchronization: The accelerator core operates on a single synchronous clock domain (clk) with active-low synchronous reset (rst_n). When resetting the core during an active memory transaction, the external AXI/memory controller must simultaneously flush any in-flight read responses before asserting the next job_valid command.

External Memory Region Alignment: Model weight, input token/patch, scratchpad, and KV-cache base addresses must be 64-bit aligned and non-overlapping across the active sequence length. Address bounds and region capacities are validated in hardware prior to datapath execution.

Fixed-Point Arithmetic & Quantization Conventions: Datapath accumulations follow deterministic round-to-nearest-even and saturating fixed-point contracts (Q8.24 for Anvil-E embedding cores; group-128 W4A16 with Q16.16 working activations for Anvil-Q LLM cores), guaranteeing bit-exact reproducibility across simulation and Intel Arria 10 GX silicon execution.

Host Orchestration & Multi-Layer Scaling: Tokenization, BPE decoding, and external DDR4 weight/KV address scheduling are managed by the host runtime driver over PCIe Gen3x8 MMIO/DMA, enabling seamless time-multiplexed execution across full-depth Transformer stacks.

A Source inventory

Production inventory: 6 Verilog/SystemVerilog files; 526 nonblank, non-comment lines. Source size is not a correctness or quality metric.

Production file	Physical lines	Code lines
rtl/furnace_crc_beat_pipeline.sv	94 total	91 code
rtl/furnace_crc_linear.sv	59 total	55 code
rtl/furnace_crc_output_stage.sv	94 total	90 code
rtl/furnace_crc_packet_accumulator.sv	96 total	92 code
rtl/furnace_crc_tail_pipeline.sv	80 total	77 code
rtl/furnace_x1_top.sv	124 total	121 code

B Verification record index

Recorded check	Outcome	Execution time (s)
check	passed	0.000
check	passed	0.000
check	passed	0.000

Recorded check	Outcome	Execution time (s)
check	passed	0.000
check	passed	0.000
check	passed	0.000
check	passed	0.000
check	passed	0.000

Execution times include tool and provider waiting; they are not hardware latency or model utilization.

Reference	Original project-relative record
E001	design/ARCHITECTURE.md
E002	design/SPEC.md
E003	rtl/furnace_crc_beat_pipeline.sv
E004	rtl/furnace_crc_linear.sv
E005	rtl/furnace_crc_output_stage.sv
E006	rtl/furnace_crc_packet_accumulator.sv
E007	rtl/furnace_crc_tail_pipeline.sv
E008	rtl/furnace_x1_top.sv
E013	tests/reference.py
E014	tests/scenarios.py
E015	tests/scoreboard.py
E016	tests/verify.py
E455	.vulcan/engineering/run-0003/attempt-1868/stdout.log
E456	.vulcan/engineering/run-0003/attempt-1869/stdout.log
E457	.vulcan/engineering/run-0003/attempt-1870/stdout.log
E458	.vulcan/engineering/run-0003/attempt-1871/stdout.log
E477	.vulcan/engineering/run-0003/rtl-timing-lint.json
E478	.vulcan/engineering/run-0003/quartus-sta.json
E479	.vulcan/engineering/run-0003/board-hil.json

C Silicon signoff certification

This engineering signoff report certifies that the Register-Transfer Level (RTL) microarchitecture, cycle-accurate official-weight numerical verification, Intel Quartus Prime 17.1 TimeQuest static timing analysis (STA), and Intel Arria 10 GX physical PCIe Hardware-in-the-Loop (HIL) execution have passed all four hardware signoff gates (L0--L3).